

Maintenance Wizard Report

Diagnosis

Advanced stage bearing failure in the CCM segment.

Root Cause

Likely fatigue failure, potentially accelerated by inadequate lubrication (contamination or starvation), excessive load, or misalignment, leading to wear and breakdown of rolling elements/raceways.

Risk Level

CRITICAL

Failure Probability

Extremely High (Imminent failure within hours to days)

Remaining Useful Life

Very Low (Unstable; can fail catastrophically at any moment)

Catastrophic Risk

HIGH - Potential for sudden segment seizure, damage to the segment housing, strand breakage, breakout, extensive secondary damage to other components, and prolonged unplanned production downtime.

Immediate Actions

[Reduce casting speed and monitor parameters (vibration, temperature, noise) intensely. If conditions worsen rapidly, consider immediate emergency shutdown.', 'Prepare all necessary tools, spares (CCM segment bearing, seals, fasteners), and personnel for urgent replacement.', 'Initiate an emergency work order for segment change out or bearing replacement.', 'Confirm availability of standby spare segment or critical bearing spares.']

Short Term Actions

[Schedule and execute the replacement of the affected CCM segment bearing (or entire segment if faster/safer) at the earliest opportunity, prioritizing machine availability.', 'Conduct a detailed visual inspection of the removed bearing, housing, and shaft for clues regarding the failure mechanism.', 'Inspect lubrication lines and adjacent bearings within the segment for any signs of similar issues, blockages, or contamination.', 'Verify proper functioning of the lubrication system to the affected area (pressure, flow, cleanliness).']

Long Term Actions

['Perform a thorough Root Cause Analysis (RCA) on the failed bearing to identify contributing factors (e.g., material defect, lubrication issues, overloading, misalignment, installation errors).', 'Review and optimize lubrication procedures, including lubricant type, quantity, frequency, and contamination control measures for all CCM segment bearings.', 'Implement or enhance continuous condition monitoring technologies (e.g., online vibration analysis, acoustic emission, thermal imaging, oil analysis) for critical CCM components.', 'Evaluate current bearing specifications and suppliers to ensure suitability for operational loads and environmental conditions.', 'Review operational parameters (casting speed, cooling rates, segment alignment checks) to identify potential stressors leading to premature bearing failure.', 'Consider upgrading to bearings with improved design, materials, or sealing if recurring failures are identified.']

Spare Procurement

['Verify current stock levels of CCM Segment Bearings and associated consumables (e.g., seals, specialized lubricants).', 'Place urgent orders to replenish any depleted stock, especially for critical spares, taking into account lead times and supplier reliability.']

Executive Summary

A CCM segment bearing is exhibiting critical failure symptoms (high vibration, increasing temperature, abnormal noise), indicating imminent catastrophic failure. This poses a HIGH risk of unplanned downtime, extensive secondary damage, and significant production loss. Immediate action is required to replace the affected bearing/segment to mitigate these risks.

Maintenance Recommendation

Urgent replacement of the failing CCM segment bearing is mandatory to prevent catastrophic failure and ensure operational continuity. This must be followed by a comprehensive Root Cause Analysis to identify systemic issues and implement preventative measures for long-term reliability.

Conclusion

The current state of the CCM segment bearing demands immediate and decisive action to prevent a major operational disruption. Proactive and thorough follow-up actions, including a robust RCA and enhanced condition monitoring, are crucial for improving overall plant reliability and minimizing future risks.